



LAB 05-CONFIGURING POP3- IMAP-MAPI-HTTP



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Exercise 1 – Configuring POP3 Services

Objective: Configure and test **POP3** protocol in Exchange 2019 using **GUI** and **PowerShell**

Tasks:

Part 1: Enable and Configure POP3 in Exchange Admin Center (EAC)

1. Open **Exchange Admin Center (EAC)** and go to **servers** → **servers**.
2. Double-click your Exchange server and select the **POP3 protocol**.
3. What **TCP port** is used and what **TCP SSL port** is used?
POP3 uses TCP port 110, and POP3 over SSL uses TCP port 995.
4. Set the **Authenticated time-out** to **10 minutes**.
5. Limit the number of maximum connections to **400**, per single IP to **200**, and per single user to **3**.
6. Save your settings.

The screenshot shows the Exchange Server configuration page for AD15. The left sidebar lists various settings: general, databases and database availability groups, POP3 (selected), IMAP4, unified messaging, DNS lookups, transport limits, transport logs, and Outlook Anywhere. The main content area displays the POP3 configuration. It includes a table for 'LOCAL IP ADDRESSES' with two rows: '(All available IPv6)' and '(All available IPv4)', both with a 'PORT' of 110. Below this is a section for 'Secure Sockets Layer (SSL) connections' with a '+ -' button. Another table for 'LOCAL IP ADDRESSES' shows '(All available IPv6)' and '(All available IPv4)' with a 'PORT' of 995. The 'Time-out settings' section includes 'Authenticated time-out (seconds):' set to 600 and 'Unauthenticated time-out (seconds):' set to 60. The 'Connection limits' section includes 'Maximum connections:' set to 400, 'Maximum connections from a single IP address:' set to 200, 'Maximum connections from a single user:' set to 3, and 'Maximum command size (bytes):' set to 512.

Part 2: Configure POP3 using Exchange Management Shell (EMS)

1. Start the **POP3** and **POP3BE** services and set them to **automatic start**.

```
[PS] C:\Windows\system32>Start-Service -Service MExchangePOP3
[PS] C:\Windows\system32>Start-Service -Service MExchangePOP3BE
[PS] C:\Windows\system32>Get-Service -Servicename MExchangePOP3, MExchangePOP3BE

Status      Name                DisplayName
-----
Running     MExchangePOP3       Microsoft Exchange POP3
Running     MExchangePOP3BE     Microsoft Exchange POP3 Backend

[PS] C:\Windows\system32>Set-Service MExchangePOP3 -StartupType Automatic
[PS] C:\Windows\system32>Set-Service MExchangePOP3BE -StartupType Automatic
[PS] C:\Windows\system32>Get-Service MExchangePOP3, MExchangePOP3BE | Select Name, Status, StartType

Name                Status StartType
-----
MExchangePOP3       Running Automatic
MExchangePOP3BE     Running Automatic
```

2. Verify POP3 settings using **Get-PopSettings**.

```

ad15 x  Ex-Client15 x
Machine: ad15.domain15.local
[PS] C:\Windows\system32>Get-PopSettings -Server "AD15" | fl

RunspaceId      : bfbe8a12-fd35-4a1b-8262-2d47e150c841
Name            : 1
ProtocolName     : POP3
MaxCommandSize   : 512
MessageRetrievalSortOrder : Ascending
InsecurelyAcceptedBindings : {[::]:110, 0.0.0.0:110}
SSLBindings      : {[::]:995, 0.0.0.0:995}
InternalConnectionSettings : {ad15.domain15.local:995/SSL, ad15.domain15.local:110/TLS}
ExternalConnectionSettings : {}
X509CertificateName : ad15
Banner           : The Microsoft Exchange POP3 service is ready.
LoginType        : SecureLogin
AuthenticatedConnectionTimeout : 00:10:00
PreAuthenticatedConnectionTimeout : 00:01:00
MaxConnections   : 400
MaxConnectionsFromSingleIP : 200
MaxConnectionsPerUser : 3
MessageRetrievalMimeFormat : BestBodyFormat
ProxyTargetPort   : 1995
CalendarItemRetrievalOption : iCalendar
OutServerVer1     : 
EnableBasicAuth : False
LiveIdBasicAuthReplacement : False
SuppressReadReceipt : False
ProtocolLogEnabled : False
EnforceCertificateErrors : False
LogFilePath       : C:\Program Files\Microsoft\Exchange Server\V15\Logging\Pop3
LogFileMailboxOverSettings : hourly
LogPerFileSizeQuota : 0 B (0 bytes)
ExtendedProtectionPolicy : None
EnableSASLAndMTLMAuth : True
Server            : AD15
AdminDisplayName  : 
ExchangeVersion   : 0.10 (14.0.100.0)
DistinguishedName : CN=1,CN=POP3,CN=Protocols,CN=AD15,CN=Servers,CN=Exchange Administrative Group (FYDIBOHF23SPDLT),CN=Administrative
  
```

3. Configure the login type to **SecureLogin**.

```

ad15 x  Ex-Client15 x
Machine: ad15.domain15.local
[PS] C:\Windows\system32>Set-PopSettings -LoginType "SecureLogin"
WARNING: The command completed successfully but no settings of 'AD15\1' have been modified.
[PS] C:\Windows\system32>_
  
```

4. Enable POP3 access for all users and verify it is enabled for the user created in Lab 3 exercise 1 step 6.

```

ad15 x  Ex-Client15 x
Machine: ad15.domain15.local
[PS] C:\Windows\system32>Get-Mailbox | Set-CASMailbox -PopEnabled:$true
[PS] C:\Windows\system32>Get-CASMailbox -Identity cliu | fl Name,PopEnabled

Name      : Chang Liu
PopEnabled : True
  
```

5. Restart the **POP3** and **POP3BE** services.

```
[PS] C:\Windows\system32>Restart-Service MExchangePOP3
WARNING: Waiting for service 'Microsoft Exchange POP3 (MExchangePOP3)' to start...
WARNING: Waiting for service 'Microsoft Exchange POP3 (MExchangePOP3)' to start...
WARNING: Waiting for service 'Microsoft Exchange POP3 (MExchangePOP3)' to start...
WARNING: Waiting for service 'Microsoft Exchange POP3 (MExchangePOP3)' to start...
WARNING: Waiting for service 'Microsoft Exchange POP3 (MExchangePOP3)' to start...
[PS] C:\Windows\system32>
[PS] C:\Windows\system32>Restart-Service MExchangePOP3BE
WARNING: Waiting for service 'Microsoft Exchange POP3 Backend (MExchangePOP3BE)' to start...
WARNING: Waiting for service 'Microsoft Exchange POP3 Backend (MExchangePOP3BE)' to start...
WARNING: Waiting for service 'Microsoft Exchange POP3 Backend (MExchangePOP3BE)' to start...
WARNING: Waiting for service 'Microsoft Exchange POP3 Backend (MExchangePOP3BE)' to start...
WARNING: Waiting for service 'Microsoft Exchange POP3 Backend (MExchangePOP3BE)' to start...
[PS] C:\Windows\system32>
[PS] C:\Windows\system32>Get-Service MExchangePOP3, MExchangePOP3BE | Select Name, Status, StartType
```

Name	Status	StartType
MExchangePOP3	Running	Automatic
MExchangePOP3BE	Running	Automatic

Exercise 2 – Configuring IMAP4 Services

Objective: Configure and test **IMAP4** protocol in Exchange 2019 using **GUI** and **PowerShell**

Tasks:

Part 1: Enable and Configure IMAP4 in Exchange Admin Center (EAC)

1. Open **Exchange Admin Center (EAC)** and go to **servers** → **servers**.
2. Double-click your Exchange server and select the **IMAP4 protocol**.
3. What **TCP port** is used and what **TCP SSL port** is used?
IMAP4 uses TCP port 143, and IMAP4 over SSL uses TCP port 993.
4. Set the connection timeout to **15 minutes**.
5. Limit the number of maximum connections to **600**, per single IP to **300**, and per single user to **4**.
6. Save your settings.

ad15 x Ex-Client15 x

Exchange Server - Profile 1 - Microsoft Edge

https://ad15.domain15.local/ecp/Servers/EditServer.aspx?pwdcid=8&ReturnObjectType=1&id=cdce40ba-7392-4527-a326-6202e525a270

AD15

general

databases and database availability groups

POP3

IMAP4

unified messaging

DNS lookups

transport limits

transport logs

Outlook Anywhere

LOCAL IP ADDRESSES	PORT
(All available IPv6)	143
(All available IPv4)	143

Secure Sockets Layer (SSL) connections:

+ -

LOCAL IP ADDRESSES	PORT
(All available IPv6)	993
(All available IPv4)	993

Time-out settings

Authenticated time-out (seconds):

900

Unauthenticated time-out (seconds):

900

Connection limits

Maximum connections:

600

Maximum connections from a single IP address:

300

Maximum connections from a single user:

4

Maximum command size (bytes):

10240

Part 2: Configure IMAP4 using Exchange Management Shell (EMS)

1. Start the **IMAP4** and **IMAP4BE** services and set them to **automatic start**.

```
Machine: ad15.domain15.local

[PS] C:\Windows\system32>Start-Service -Service MExchangeIMAP4
WARNING: Waiting for service 'Microsoft Exchange IMAP4 (MExchangeIMAP4)' to start...
WARNING: Waiting for service 'Microsoft Exchange IMAP4 (MExchangeIMAP4)' to start...
WARNING: Waiting for service 'Microsoft Exchange IMAP4 (MExchangeIMAP4)' to start...
WARNING: Waiting for service 'Microsoft Exchange IMAP4 (MExchangeIMAP4)' to start...
WARNING: Waiting for service 'Microsoft Exchange IMAP4 (MExchangeIMAP4)' to start...
[PS] C:\Windows\system32>Start-Service -Service MExchangeIMAP4BE
WARNING: Waiting for service 'Microsoft Exchange IMAP4 Backend (MExchangeIMAP4BE)' to start...
WARNING: Waiting for service 'Microsoft Exchange IMAP4 Backend (MExchangeIMAP4BE)' to start...
WARNING: Waiting for service 'Microsoft Exchange IMAP4 Backend (MExchangeIMAP4BE)' to start...
[PS] C:\Windows\system32>Get-Service -Servicename MExchangeIMAP4, MExchangeIMAP4BE

Status Name DisplayName
-----
Running MExchangeIMAP4 Microsoft Exchange IMAP4
Running MExchangeIMAP4BE Microsoft Exchange IMAP4 Backend

[PS] C:\Windows\system32>Set-Service MExchangeIMAP4 -StartupType Automatic
[PS] C:\Windows\system32>Set-Service MExchangeIMAP4BE -StartupType Automatic
[PS] C:\Windows\system32>Get-Service MExchangeIMAP4, MExchangeIMAP4BE | Select Name, Status, StartType

Name Status StartType
-----
MExchangeIMAP4 Running Automatic
MExchangeIMAP4BE Running Automatic
```


2. Verify IMAP4 settings using **Get-ImapSettings**.

```

PS C:\Windows\system32>Get-ImapSettings -Server "AD15" | fl

RunspaceId      : bf8e8a12-fd35-4a1b-8262-2d47e158c843
ProtocolName     : IMAP4
Name            : 1
MaxCommandSize  : 10240
ShowHiddenFoldersEnabled : False
ImapEncryptedOrTLSBindings : {[::]:143, 0.0.0.0:143}
SSLBindings      : {[::]:993, 0.0.0.0:993}
InternalConnectionSettings : {ad15.domain15.local:993:SSL, ad15.domain15.local:143:TLS}
ExternalConnectionSettings : {}
X509CertificateName : ad15
Banner           : The Microsoft Exchange IMAP4 service is ready.
LoginType        : SecureLogin
AuthenticatedConnectionTimeout : 00:15:00
PreAuthenticatedConnectionTimeout : 00:15:00
MaxConnections  : 500
MaxConnectionsPerSingleIP : 300
MaxConnectionsPerUser : 4
MessageRetrievalTimeFormat : BriefBodyFormat
ProxyTargetPort  : 993
CalendarItemRetrievalOption : iCalendar
OwsServerUrl     : 
EnableExactMatchSize : False
LiveIdBasicAuthReplacement : False
SuppressReceipt : False
ProtocolLogEnabled : False
EnforceCertificateErrors : False
LogFileLocation  : C:\Program Files\Microsoft\Exchange Server\V15\Logging\Imap4
LogFileRolloverSettings : Hourly
LogPerFileSizeQuota : 0 MB (0 bytes)
ExtendedProtectionPolicy : None
EnableSSAPIAndNTLMAuth : True
Server           : AD15
AdminDisplayNone : 
ExchangeVersion  : 0.10 (14.0.100.0)
DistinguishedName : CN=1,CN=IMAP4,CN=Protocols,CN=AD15,CN=Servers,CN=Exchange Administrative Group (FYDIBOHF23SPDLT),CN=Administrative
  
```

3. Limit the number of **maximum connections** to 500.

```

PS C:\Windows\system32>Set-ImapSettings -Server "AD15" -MaxConnections 500
WARNING: Changes to IMAP4 settings will only take effect after all Microsoft Exchange IMAP4 services are restarted on server AD15.
PS C:\Windows\system32>
  
```

4. Enable **IMAP4** access for **all mailboxes** and verify it is enabled for the **user created in Lab 3 exercise 1 step 6**.

```

PS C:\Windows\system32>Get-Mailbox | Set-CASMailbox -ImapEnabled:$true
PS C:\Windows\system32>Get-CASMailbox -Identity cliu | fl Name,ImapEnabled

Name      : Chang Liu
ImapEnabled : True
  
```

5. Restart the **IMAP4** and **IMAP4BE** services.

```

PS C:\Windows\system32>Restart-Service MExchangeIMAP4
WARNING: Waiting for service 'Microsoft Exchange IMAP4 (MExchangeIMAP4)' to start...
PS C:\Windows\system32>Restart-Service MExchangeIMAP4BE
WARNING: Waiting for service 'Microsoft Exchange IMAP4 Backend (MExchangeIMAP4BE)' to start...
WARNING: Waiting for service 'Microsoft Exchange IMAP4 Backend (MExchangeIMAP4BE)' to start...
WARNING: Waiting for service 'Microsoft Exchange IMAP4 Backend (MExchangeIMAP4BE)' to start...
PS C:\Windows\system32>
PS C:\Windows\system32>Get-Service MExchangeIMAP4, MExchangeIMAP4BE | Select Name, Status, StartType

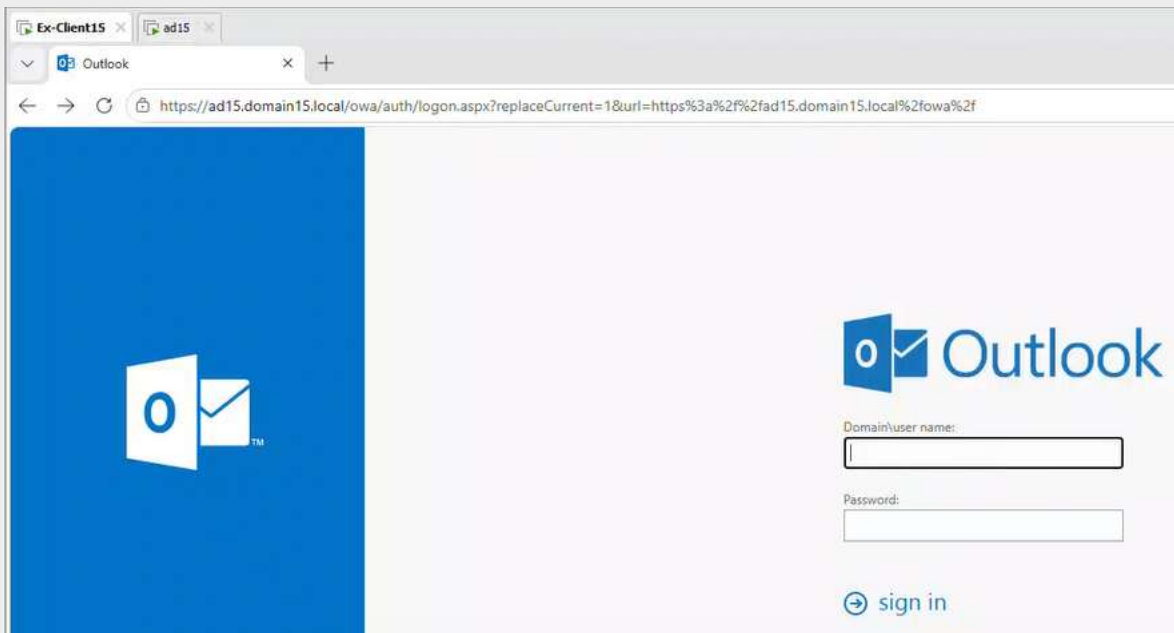
Name      Status StartType
-----
MExchangeIMAP4 Running Automatic
MExchangeIMAP4BE Running Automatic
  
```

Exercise 3 – Testing POP3 and IMAP4 on Windows 11

Objective: Configure and test **POP3** and **IMAP4** on a **domain-joined Windows 11** client

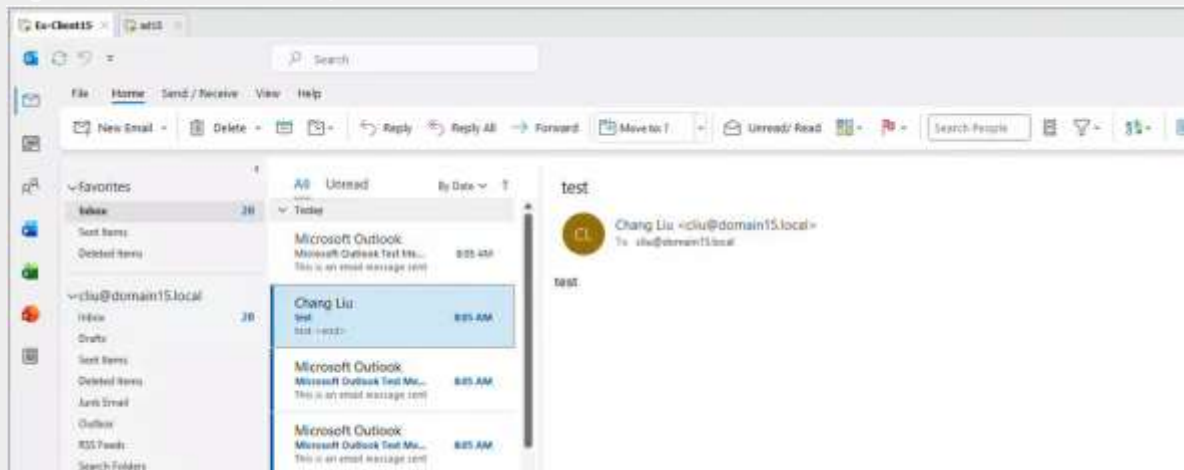
Tasks:

1. Using **Lab 05A-Export Exchange Certificate**, export the **Exchange Server certificate** on your **Windows 11** machine.

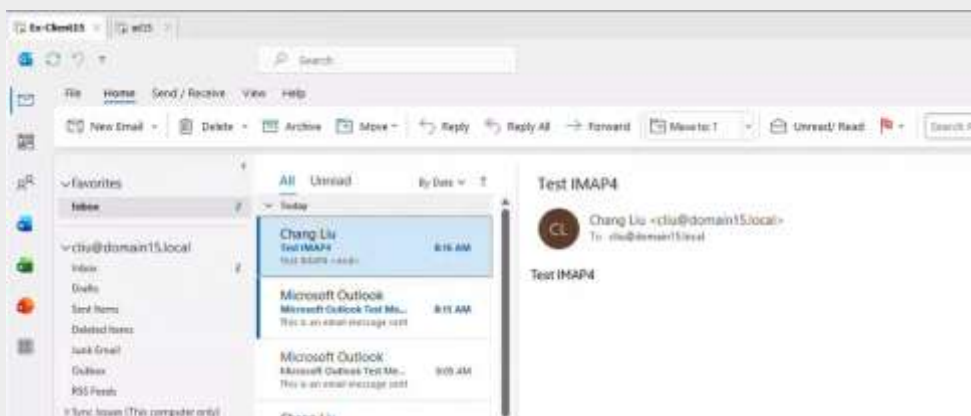


2. Using **Lab 05B-Configure Outlook Classic -POP3-IMAP4**, do the following two tasks:
 - Configure **Outlook Classic** to connect to **Exchange** using **POP3** and verify email download.





- Reconfigure **Outlook Classic** to connect to **Exchange** using **IMAP4** and verify synchronization with the server.



Exercise 4 – Configuring and testing MAPI over HTTP

Objective: Enable and verify **MAPI/HTTP** connectivity for Outlook clients

Tasks:

On the Exchange server:

1. Check **MAPI/HTTP** status at the **organization level** using **PowerShell**.

Lab 05 - Configure POP3, IMAP and MAPI/HTTP services

```
ad15 x Ex-Client15 x
Machine: ad15.domain15.local
[PS] C:\Windows\system32>Get-OrganizationConfig | Format-List -Property MapiHttpEnabled

MapiHttpEnabled : True
```

2. Enable **MAPI/HTTP** protocol for all mailboxes.

```
ad15 x Ex-Client15 x
Machine: ad15.domain15.local
[PS] C:\Windows\system32>Get-CASMailbox | Set-CASMailbox -MAPIEnabled $true
[PS] C:\Windows\system32>Get-CASMailbox | Select-Object Name,MAPIEnabled

Name                                     MAPIEnabled
----
Administrator                           True
Chang Liu                               True
DiscoverySearchMailbox {D919BA05-46A6-415F-80AD-7E093348B852} True
Elon Musk                                True
```

3. Configure **MAPI virtual directory** with external URL.

```
ad15 x Ex-Client15 x
Machine: ad15.domain15.local
[PS] C:\Windows\system32>Set-MapiVirtualDirectory -Identity "mapi (Default Web Site)" -ExternalUrl https://mail.domain15.ca/mapi -IISAuthenticationMethods Negotiate
[PS] C:\Windows\system32>Get-MapiVirtualDirectory | Select-Object Server,Name,ExternalUrl

Server Name                                     ExternalUrl
-----
AD15 mapi (Default Web Site) https://mail.domain15.ca/mapi
```

4. Verify **MAPI internal** and **external URL**.

```
ad15 x Ex-Client15 x
Machine: ad15.domain15.local
[PS] C:\Windows\system32>Get-MapiVirtualDirectory -Identity "mapi (Default Web Site)"

Name                                     Server InternalUrl                                     ExternalUrl
----
mapi (Default Web Site) AD15 https://ad15.domain15.local/mapi https://mail.domain15.ca/mapi
```

5. Test the **MAPI/HTTP** configuration.

```
ad15 x Ex-Client15 x
Machine: ad15.domain15.local
[PS] C:\Windows\system32>Test-OutlookConnectivity -RunFromServerId ad15 -ProbeIdentity OutlookMapiHttpSelfTestProbe

MonitorIdentity                                     StartTime                                     EndTime                                     Result                                     Error                                     Exception                                     Poisoned
-----
OutlookMapiHttp.Protocol\OutlookMapiH... 6/9/2026 3:2... 6/9/2026 3:2... Succeeded                                     0
```

6. Using **Lab 05C-Configure Outlook Classic -MAPI/HTTP**, do the following two tasks:

- Reconfigure **Outlook Classic** to connect to **Exchange** using **MAPI/HTTP** and verify synchronization with the server.

Lab 05 - Configure POP3, IMAP and MAPI/HTTP services

